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The HUMAN Digest 1.4

Lake Forest College's AI newsletter, courtesy of the HUMAN grant



Image created in Midjourney by David Sanchez-Burr

Greetings. In this issue you will find:

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HUMAN Digest 1.4

Welcome to the fourth issue of the HUMAN Digest! It's hard to ignore just how fast AI is making its way into classrooms and transforming the way we teach and learn.

Tools like ChatGPT have become more than just novelties—they're reshaping how students interact with course material and how educators deliver it (to the frustration of many). All of this comes with a responsibility to rethink how we handle assessments, maintain academic integrity, and ensure that every student has access to these tools.

In this issue, we'll look at the intersection of AI and the humanities, featuring exhibitions and initiatives that explore AI's impact on creativity, education, and cultural expression. We'll also cover recent developments in AI research and technology, including Nobel Prize awards to key figures in the field. Additionally, we'll touch on various perspectives on the future of AI, from optimistic visions of its potential to concerns about regulation and societal impact.

We hope to give you insights into how this rapidly changing technology is shaping higher education, while keeping a close eye on the implications for students, educators, and the world at large. The HUMAN Digest welcomes feedback at kee@lakeforest.edu

Justin Kee, editor-in-chief

Davis Schneiderman, Principal Investigator, [HUMAN](#) grant

AI and the Humanities

- The University of Bergen (Norway) Center for Digital Narrative (CDN) opened the second iteration of “[More than Meets AI](#),” a five-site exhibition that engages with AI in [ways](#) that challenge conventional notions of how artists and writers might respond to this tectonic shift in technology, considering for example the complexity of new forms of writing that manifest as visual art. How might we “read” a parody articulated through generated images? How will AI impact the future of motion pictures? How can AI be used to produce works that critique dominant power structures? How can artworks address the biases and inequities that are baked into AI platforms and perhaps embedded within human language itself?”

The exhibition continues the CDN’s work around electronic literature, and a seminar this past week featured artists and critics (including HUMAN PI Davis Schneiderman) presenting on aspects of digital narrative and AI as they prepare a special-issue journal publication.

- Closer to home, the [Blurring the Lines](#) exhibition at Indiana University-Bloomington exhibition explores AI’s intersection with human creativity. The exhibition explored the intersection of AI and culture, highlighting how AI can inspire new forms of creative expression. Instead of focusing on the potential downsides like deepfakes, the exhibition invited viewers to consider how AI can be a valuable tool for innovation.

Featured works included “I.A. Suzie,” which reflects on AI’s role in elder care; “Generated Love,” which “reimagines Ming Dynasty porcelain with counterfactual AI-generated imagery;” and “Self-portrait Ai-Da,” a creation of the humanoid robot artist

Ai-Da. Another fascinating piece was Megan Young's "Carry," a conversational AI that interacts with women from around the world, discussing themes like displacement and forced migration—adding voices to stories that might otherwise be unheard. Sougwen Chung's "DOUG" (Drawing Operations Unit Generation_X) offered a collaborative experience, where a robot synchronously paints alongside the artist, tracking data from an EEG headset that monitors brain activity during the process.

- MIT faculty are integrating AI into various departments [*to enhance both teaching and learning*](#). In one example, a Japanese language class used AI tools to help students improve vocabulary and grammar by comparing their work with AI-generated sentences.
- Stanford's Center for Teaching and Learning offers [*helpful advice*](#) for educators on how to adapt to AI tools in the classroom. They encourage teachers to explore how AI can both challenge and enrich traditional teaching methods. By weaving AI into assignments and class discussions, educators can guide students in understanding both the ethical and practical sides of these tools. This also highlights the importance of creating assignments that inspire students to think deeply and develop their own ideas, rather than just leaning on AI for answers.

General news

- Geoffrey Hinton, profiled in issue 1.1 of the HUMAN digest, was awarded the [Nobel Prize in Physics](#) for his work on neural networks. Read more [here](#).
- John Hopfield was also [awarded the prize](#); he invented the Hopfield network, a breakthrough that laid the groundwork for modern machine-learning technologies
- Demis Hassabis, David Baker and John M. Jumper were awarded the [Nobel Prize in Chemistry](#). Demis co-founded DeepMind in 2010, which notably created the self-learning [AlphaGo](#) model that beat the world champion, Lee Sedol, in 2016. Here is a relatively recent [interview](#) with him.
- Northwestern University's Buffett Center for Global Affairs, along with the McCormick School of Engineering, which supports the [Northwestern Security & AI Lab](#), recently hosted a [conference on AI and national security](#).
- Recent [research](#) by OpenAI explored bias in AI models based on whether or not the model knows the user's name. The paper refers to the concept of "first person fairness", outputs directed towards the user which may exhibit bias, as opposed to "third person fairness," which is studied in the context of algorithmic bias in areas such as loan approval and resume screening.

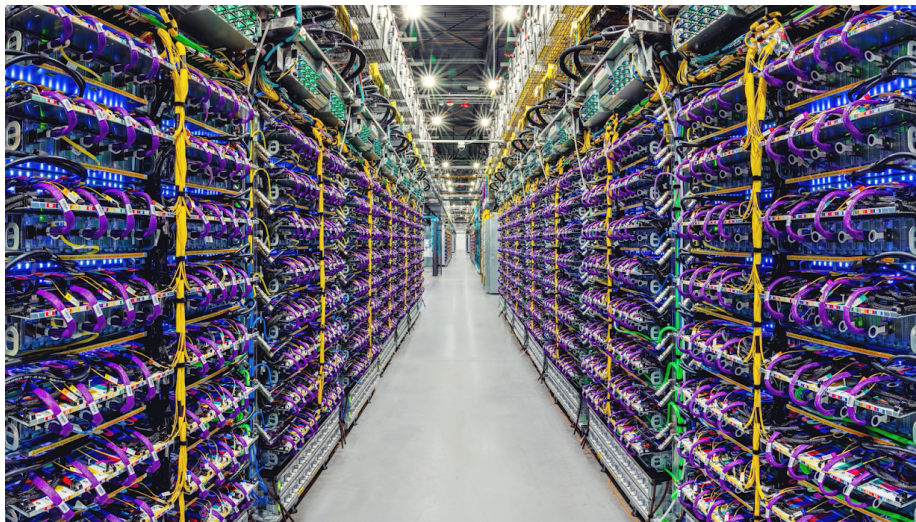


Image courtesy Google DeepMind.

Other perspectives

- Dario Amodei, of the research lab Anthropic, which produces the [Claude.ai](#) model, offered his vision for the future of AI in a recent [article](#). Titled “Machine of Loving Grace,” Dario explores how more capable AI systems will impact biology, neuroscience, the economy, war and peace and the meaning of work.
- Go deeper: For those of you who like to listen to podcast interviews, this Dwarkesh Patel [interview](#) with Amodei is quite insightful.
- In an [opinion piece](#) in Time magazine, Daniel Colson, executive director of the [AI Policy Institute](#), articulates that Artificial General Intelligence (AGI) could reshape our world—displacing jobs, revolutionizing economies, and introducing unprecedented risks like cyberattacks or worse. As other AI insiders warn, the window for regulating AGI might be closing faster than anticipated. Policymakers are urged to implement stronger regulatory frameworks and ensure government transparency before AGI systems outpace society’s ability to

manage them. Public awareness and involvement are also crucial; AGI is not merely a technical challenge, but a societal one that could have far-reaching consequences for everyone.

Of course, not everyone agrees that AGI is around the corner. Matteo Wong writes in [The Atlantic](#) (paywalled) about how AGI may be part of a complicated hype cycle. Wong argues that prominent AI executives are making ambitious predictions about AGI amid rising concerns over AI's massive energy and capital requirements. Wong suggests these bold claims may be attempts to justify unprecedented spending and maintain investor faith, rather than reflections of actual technological breakthroughs.

A few useful terms

Artificial General Intelligence (AGI) refers to a type of AI that can understand, learn, and apply intelligence across a broad range of tasks, much like a human. Unlike narrow AI, which is designed to perform specific tasks (like facial recognition or language translation), AGI has the ability to handle any cognitive task that a human can, including reasoning, problem-solving, learning from experience, and adapting to new situations. The goal of AGI is to create systems that are not limited to predefined functions but can learn and perform in a flexible, generalized manner across different domains. This represents a major leap from current AI systems, which are often highly specialized in narrow areas. Note: the term is fluid and many in the field have varying definitions of what constitutes AGI.

Overfitting In machine learning, overfitting occurs when a model learns not just the general patterns in the training data, but also the noise and specific details that don't apply to new data. Essentially, the model becomes too “fitted” to the training set, which can make it less effective at making predictions on unseen data.

Think of it like studying for a test by memorizing all the questions and answers from practice tests instead of understanding the underlying concepts—you might do well on the practice tests but struggle with the actual exam if the questions are different. A well-balanced model finds a sweet spot between underfitting (not learning enough) and overfitting, allowing it to generalize well to new data.

People to know

Demis Hassabis is a leading figure in artificial intelligence, best known as the co-founder and CEO of [DeepMind](#), an AI research lab acquired by Google in 2014. Born in London in 1976, Hassabis was a chess prodigy and earned the title of master at just 13 years old. At 17, he was co-designer and lead programmer on the video game, [Theme Park](#), released in 1994. He went on to study computer science at the University of Cambridge before completing a PhD in cognitive neuroscience at University College London. Hassabis' expertise in both AI and neuroscience led him to explore how human cognition could contribute to the development of new techniques in machine learning and artificial intelligence. Apart from his Nobel Prize, he was knighted in 2024 for his service to the government of the United Kingdom.



Image courtesy of Wikipedia.

At DeepMind, he led the development of novel AI systems, including AlphaGo, which famously defeated the world champion Go player in 2016, a milestone in AI development. Hassabis' work has been driven by the goal of creating AI that can solve complex problems across a wide range of fields, from healthcare to scientific research. His research into neural networks and reinforcement learning has positioned him as a thought leader in the AI field, advocating for the responsible development of AI technologies that can benefit society. In addition to his work at DeepMind, he has published numerous papers and remains a key voice in discussions about the ethical implications of AI.

Technical AI

A notable paper in the field of deep learning:

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

[Chain-of-thought](#) - A technique whereby LLMs can break down complex problems into sequential steps, mimicking the human reasoning processes.

Several videos to watch

Grant Sanderson of [3blue1brown](#) has a fantastic set of videos that explains the math behind computational [neural networks](#) and the [transformer architecture](#).

Just for fun

Create and converse with virtual [characters](#). These characters use large language models to process text, enabling them to engage in realistic, human-like conversations.

The McFlurry [index](#).

HUMANS IN THE LOOP.

A warm thank you to research assistants Estelle Ghinea, Kyle Moyo and Suvexa Pradhan Tuladhar for the help with this issue of the HUMAN digest.

Have a tip or suggestion for an upcoming issue of the HUMAN Digest? Write Justin Kee an email at kee@lakeforest.edu.

